CS 495: Iteration 2 Presentation

Multi-User Physiological Computing



Dr. Chris Crawford



Zara Ahmed



Trenton Ottman



Jackson Wallace



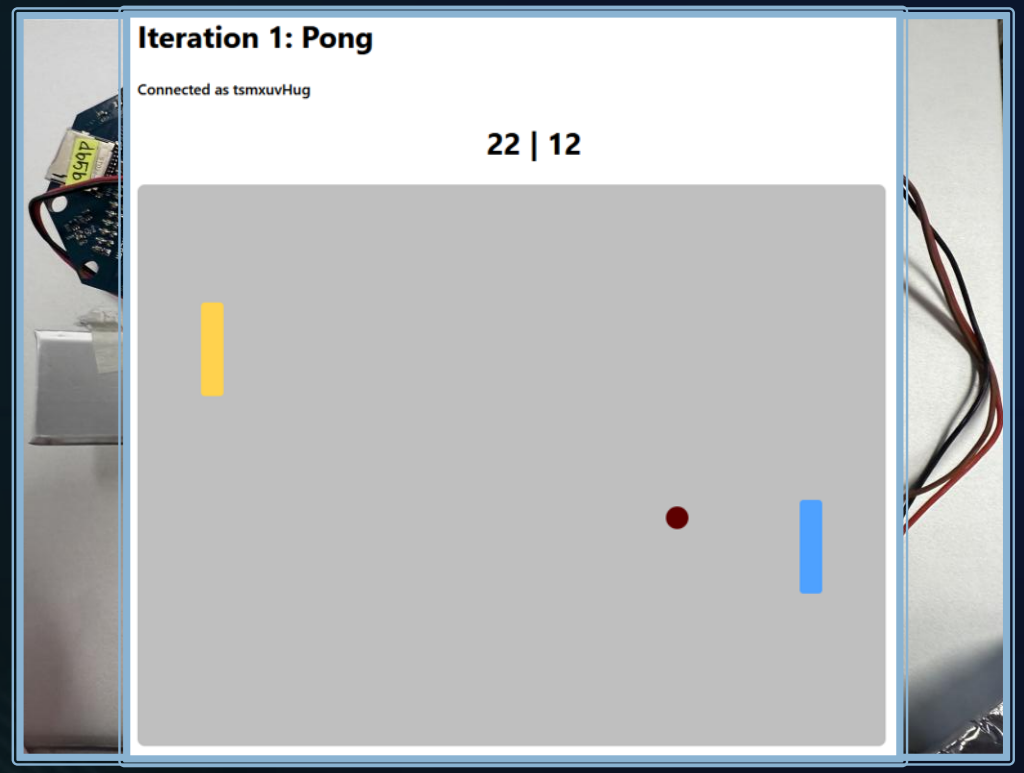
Ansel Stinnett



James Adler

Speaker:
Everyone

Project Review



- Goal: Create a multiplayer game utilizing EMG sensors.
- Iteration 1: Proof of concept multiplayer pong implementation.
- Iteration 2: MVP 8 player multiplayer game with EMG input.

Speaker:
James Adler

Planning & Actualizing

- Iteration goal: make an 8-12 player multiplayer game with sensor used for inputs
- Pick final game idea
- Make project user friendly

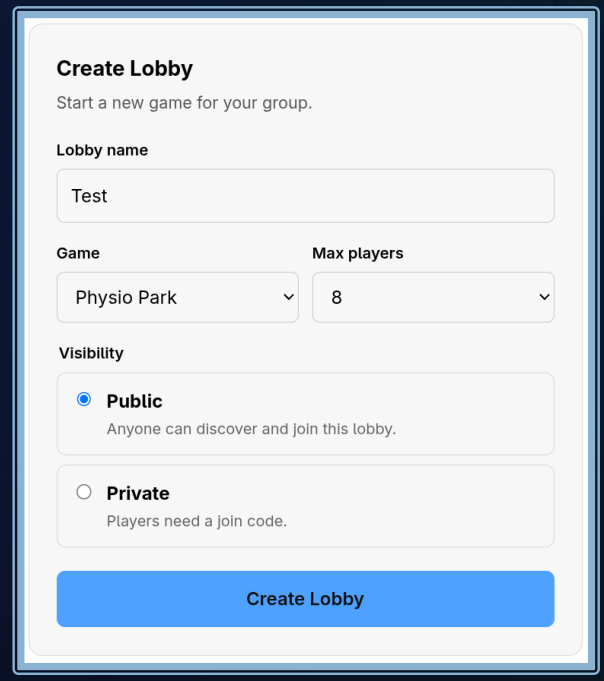
- Created lobby system
- Updated Sensor UI
- Created Physio Park game

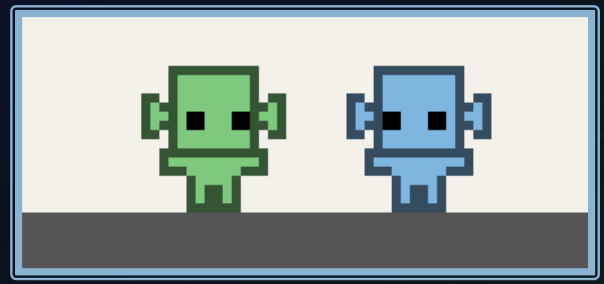


Speaker:
Jackson Wallace

Major Accomplishments

- Created a working lobby system, including public and private rooms
- Established generalized game physics
- Translated sensor inputs into player movement (jumping)
- Designed hand-drawn sprites and animations
- Built 2 playable levels





Speaker:
Trenton Ottman

Engineering & Quality

```ts
const goalPlatform = { x: 256, y: GROUND_TOP - 30, width: 64, height: 6 } as const;
const goalArea = {
    x: goalPlatform.x,
    y: goalPlatform.y - 30,
    width: goalPlatform.width,
    height: 30,
} as const;

/** Collision shapes for the introductory stepping-platform level. */
export const LEVEL_1_COLLISION: PhysioParkLevelCollision = {
    width: LEVEL_WIDTH,
    height: LEVEL_HEIGHT,
    solids: [
        { x: 0, y: GROUND_TOP, width: 100, height: 24 },
        { x: 120, y: 138, width: 24, height: 6 },
        { x: 164, y: 126, width: 24, height: 6 },
        { x: 208, y: GROUND_TOP, width: 112, height: 24 },
```

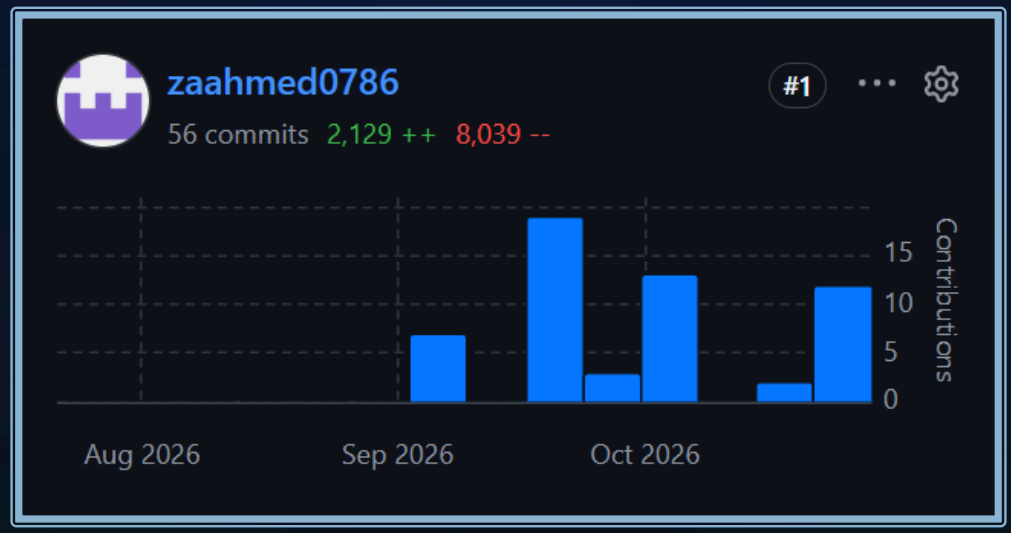
- #107: Added Sensor Input

- docs: created plan for sensor interface
  #80 · by LocandLoad was merged 2 weeks ago Collaborator · Approved
- Pbi 15 Handle player disconnects
  #77 · by LocandLoad was merged 2 weeks ago Collaborator · 3 tasks done · Approved
- Create game-resolution-etc.md
  #75 · by zaahmed0786 was merged 2 weeks ago Collaborator · Approved
- PBI-10 and PBI-11 Join Codes and Lobby Browser
  #73 · by TrentonOttman was merged 3 weeks ago Owner · 3 tasks done · Approved
- added character sprites to docs/planning
  #72 · by LocandLoad was merged 3 weeks ago Collaborator · Approved
- Setup 04: Initialize Physio Park
  #71 · by JamesAdler32 was merged 3 weeks ago Collaborator · 2 tasks done · Approved

Speaker:
Zara Ahmed

Team Contributions - Zara

- Documentation for Level 1 & general game design
- Added a player naming system
- Made tile sets & level editor setup



Speaker:
Zara Ahmed

Team Contributions - James

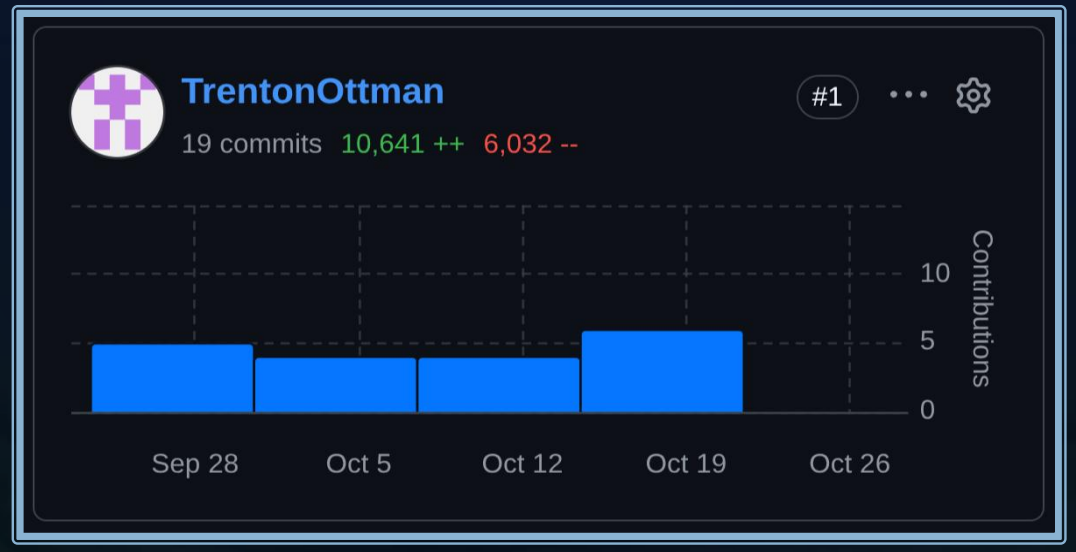
- Partially generalized physics implementation
- Physio Park setup, resolution, and scaling
- Asset loading and some animations



Speaker:
James Adler

Team Contributions - Trenton

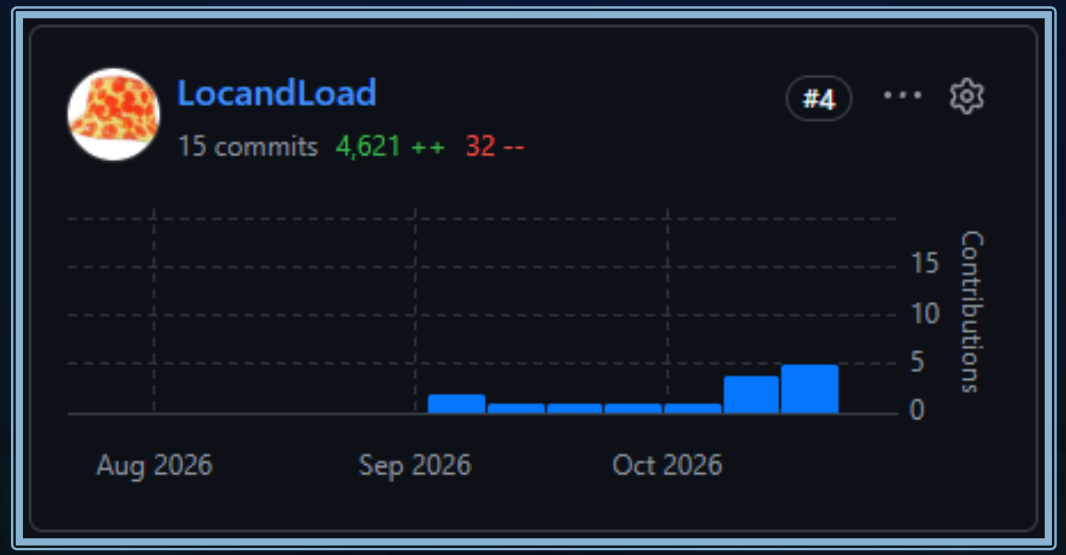
- Public and private lobbies, lobby waiting rooms, host controls, and starting a game
- Level completions and transitions
- Level 2 design, sprites, and implementation



Speaker:
Trenton Ottman

Team Contributions - Jackson

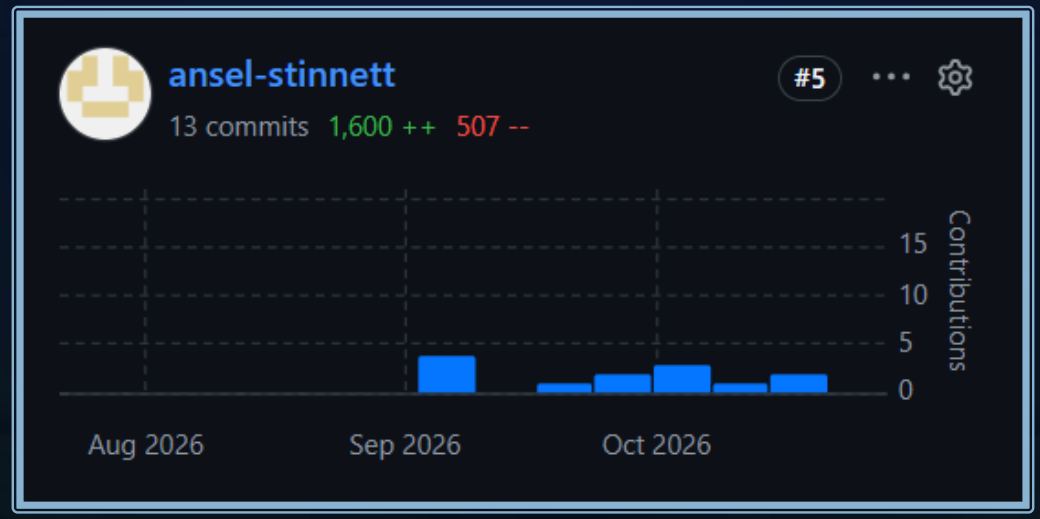
- Created sprites for Physio-Park
- Worked on using the sensor as an input for the user
- Worked on allowing users to rejoin on disconnect



Speaker:
Jackson Wallace

Team Contributions – Ansel

- Reviewed Game engine
- Built platformer physics and stacking
- Connected the physics to the multiplayer server for controlling movement
- Groundwork for client-side prediction



Speaker:
Ansel Stinnett

Lessons Learned & Upcoming

- **Iteration 3:**
  - Client-side prediction (for smoother movement)
  - Add more levels
  - Improve custom parallax
  - Play-test with more (8) players.
- **Iteration 2:**
  - Phaser/Typescript technical learning
  - Implement physics
- **Iteration 1:**
  - Lobby and room system from Iteration 1 carried straight into Physio Park which allowed for focus on gameplay
- PBI: helpful to work in parallel, but coordinating early can help merge order in the future.
- Running the physics on the server keeps player in sync & EMG input needed tuning

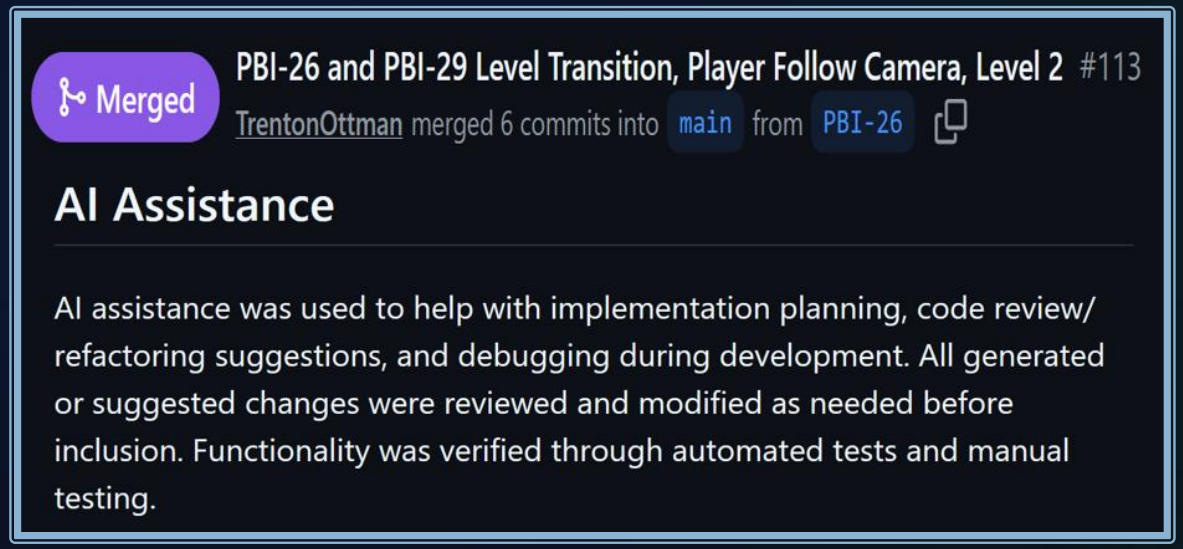




Speaker:
Ansel Stinnett

AI Usage

- Documentation
- Prototyping and code assistance
- Debugging



Speaker:
James Adler

Questions?

